

ECE157C HW4 Report - Orchestrated Agent Design

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1. Introduction

This project develops an orchestrated AI analytics system that enables users to analyze financial datasets using natural-language questions. The system combines Large Language Models (LLMs) with deterministic data analysis tools to provide accurate, explainable, and data-driven insights. The dataset consists of company financial information from 2014 to 2018, including revenue, profitability, cash flow, valuation ratios, and sector classifications.

The system is built around a multi-agent architecture. A Planner Agent interprets user questions and generates structured analysis plans. Depending on the task, requests are routed to either an Analytics Agent for dataset-based analysis or a Web Search Agent for external information. The Analytics Agent performs computations, aggregations, comparisons, and visualizations using pandas and Plotly. A Memory component maintains conversation context and stores intermediate results.

To improve reliability, a Validator Agent independently reviews the generated outputs and compares them against the original user question. The validator checks analytical correctness, completeness, numerical consistency, and whether the requested analysis was fully addressed. If issues are detected, the system can recommend additional reasoning or re-execution.

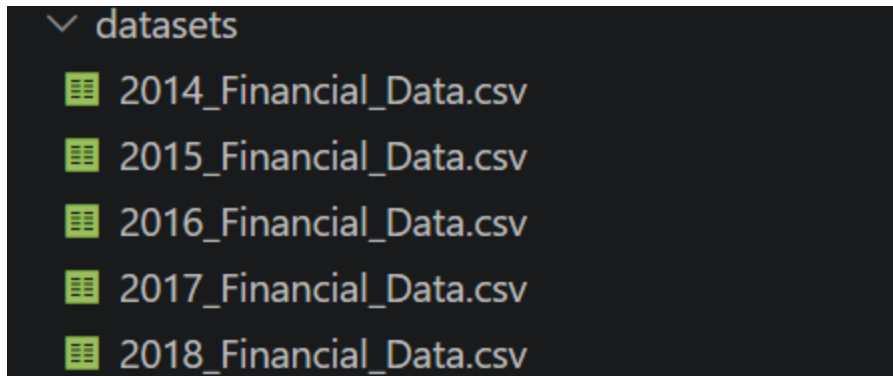
The final system supports sector and company ranking, financial strength analysis, valuation screening, correlation analysis, cross-year comparisons, and external financial knowledge queries. By separating planning, execution, and validation into specialized agents, the system provides a more robust and trustworthy analytics workflow than traditional single-agent approaches.

2. Dataset

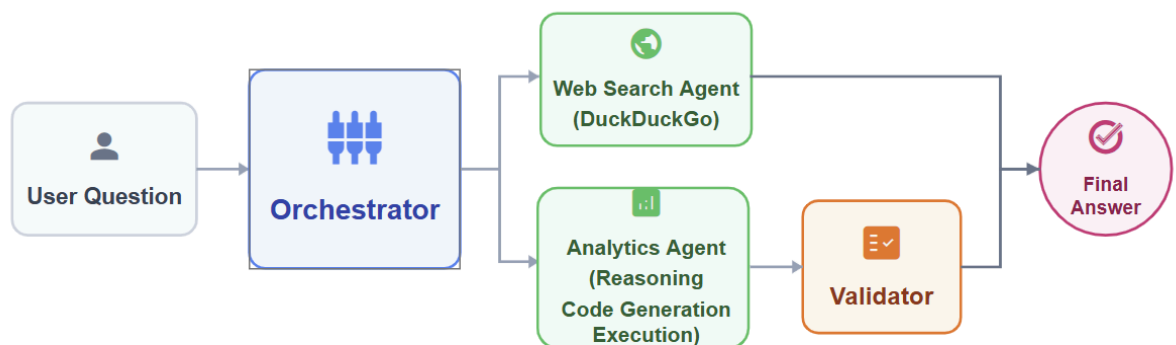
This system uses the financial data sourced from the Kaggle repository titled "200+ Financial Indicators of US Stocks (2014–2018)."

Link to the dataset:

<https://www.kaggle.com/datasets/cnic92/200-financial-indicators-of-us-stocks-20142018>



3. System Architecture



4. Orchestrator Design

The Orchestrator serves as the central controller of the system and coordinates interactions between all agents. When a user submits a question, the Orchestrator first sends it to the Planner Agent, which generates a structured plan describing the type of analysis required. Based on this plan, the Orchestrator routes the request either to the Analytics Agent for dataset-based computations or to the Web Search Agent for external information retrieval.

After the selected agent produces an answer, the Orchestrator forwards the results to the Validator Agent. The validator independently checks the correctness, completeness, and consistency of the analysis. If issues are detected, the Orchestrator can trigger a retry or provide validation feedback to the user. Throughout the process, the Orchestrator also interacts with the Memory component to store conversation history and analysis results, enabling multi-turn interactions.

By separating planning, execution, validation, and memory management, the Orchestrator ensures that each component focuses on a specialized task while maintaining a coordinated workflow for reliable analytics.

5. Analytics Agent Design

The Analytics Agent is responsible for performing dataset-driven analysis using the financial CSV files. It follows an iterative workflow that converts a structured analysis plan into analytical results, visualizations, and natural-language explanations. This design separates reasoning from execution, allowing the agent to perform reliable calculations while maintaining transparency through an execution trace.

The workflow consists of five main steps:

- a. **Inspect Dataset**
The agent first loads the available CSV files and examines their structure, including the number of rows, columns, available years, and data types. This step helps determine which datasets are relevant to the user's request.
- b. **Select Relevant Columns**
Based on the analysis plan, the agent identifies the financial metrics required to answer the question, such as Revenue, Net Income, Free Cash Flow, ROE, ROIC, or Debt-to-Equity Ratio. For multi-year analyses, the agent also selects the appropriate yearly datasets.
- c. **Clean and Analyze Data**
The selected columns are converted to numeric formats, missing values are handled, and aggregation operations such as grouping, filtering, ranking, correlation analysis, or cross-year comparisons are performed. Depending on the question, the agent may compute sector averages, growth rates, valuation scores, or composite financial strength scores.
- d. **Generate Plotly Visualization**
To improve interpretability, the agent creates interactive Plotly charts, such as bar charts, scatter plots, or trend visualizations. These visualizations provide graphical evidence supporting the analytical findings.
- e. **Produce Final Answer**
Finally, the agent converts the analytical results into a natural-language response that summarizes the key findings, includes numerical evidence, and references the generated visualization. The execution trace is also recorded to document each step of the analysis process.

This iterative loop enables the Analytics Agent to transform raw financial data into meaningful insights while maintaining explainability and reproducibility.

6. Validator Agent Design

The Validator Agent acts as an independent reasoning and checking component that evaluates the outputs generated by the Analytics Agent. Rather than simply verifying that the code executed successfully, the validator examines whether the analysis correctly answers the user's question and whether the results are reliable and complete.

The validator performs several checks:

- a. **Completeness Check**
It determines whether the generated answer fully addresses the user's request. For example, if a question asks for the financially strongest sectors, the validator verifies that multiple financial metrics were considered rather than relying on a single metric such as revenue.
- b. **Numerical Consistency Check**
The validator inspects the analytical results for suspicious or unrealistic values that may indicate data quality issues or calculation errors. It also verifies that ranking directions (e.g., highest vs. lowest, strongest vs. weakest) are consistent with the user's question.
- c. **Correct File Usage Check**
For multi-year analyses, the validator ensures that the appropriate datasets were used. For example, a question comparing 2014 and 2018 should involve both yearly files rather than a single dataset. This helps prevent incorrect conclusions caused by incomplete data selection.
- d. **Reasoning and Validation Comparison**
The validator independently analyzes the question and compares its expectations with the outputs generated by the Analytics Agent. If important metrics, years, or analytical steps are missing, the validator flags the result as incomplete or suspicious.
- e. **Retry Recommendation**
When inconsistencies or missing analyses are detected, the validator can recommend a retry. This may involve re-running the analysis with corrected parameters, additional datasets, or a more appropriate analytical method.

By acting as a separate verification stage, the Validator Agent improves the reliability, transparency, and trustworthiness of the overall analytics system. It helps ensure that the final answer is not only generated successfully but is also logically consistent and sufficiently supported by the data.

7. Six representative examples

First 2 examples: generic-domain Questions:

1. What is retrieval -augmented generation?
2. What is NVIDIA's primary business segment?

The image displays three sequential screenshots of the FinSight Orchestrator web application, illustrating the Retrieval-Augmented Generation (RAG) workflow. The interface is titled "FinSight Orchestrator" and "Multi-agent AI analytics system for financial stock indicators".

Screenshot 1: Query 1
User: What is retrieval-augmented generation?
Assistant: Retrieval-augmented generation (RAG) is a machine learning approach that combines retrieval-based methods with generative models to enhance the quality and relevance of generated text. It retrieves relevant information from a database or knowledge source and uses that information to inform and improve the text generation process.
Sources: [RAG Paper on arXiv](https://arxiv.org/abs/2005.11401)
Execution / Validation Trace:

```
[{"step": "query_rewrite", "original_question": "What is retrieval-augmented generation?", "query_variants_used": [{"query": "What is retrieval-augmented generation?"}], "step": "duckduckgo_search"}
```

Screenshot 2: Query 2
User: What is NVIDIA's primary business segment?
Assistant: NVIDIA's primary business segment is Graphics Processing Units (GPUs) for gaming and professional markets, along with a significant focus on artificial intelligence (AI) computing.
Sources: [NVIDIA Official Site](http://www.nvidia.com/page/home.html), [NVIDIA Wikipedia](https://en.wikipedia.org/wiki/Nvidia)
Execution / Validation Trace:

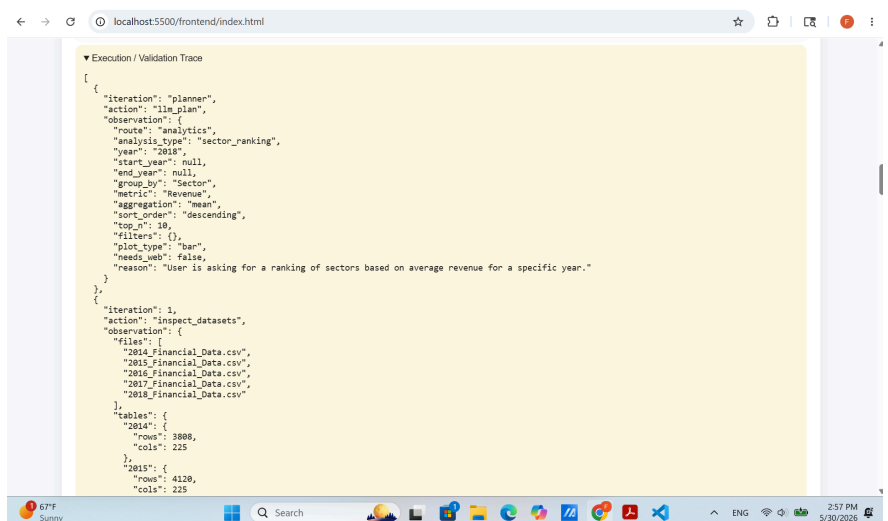
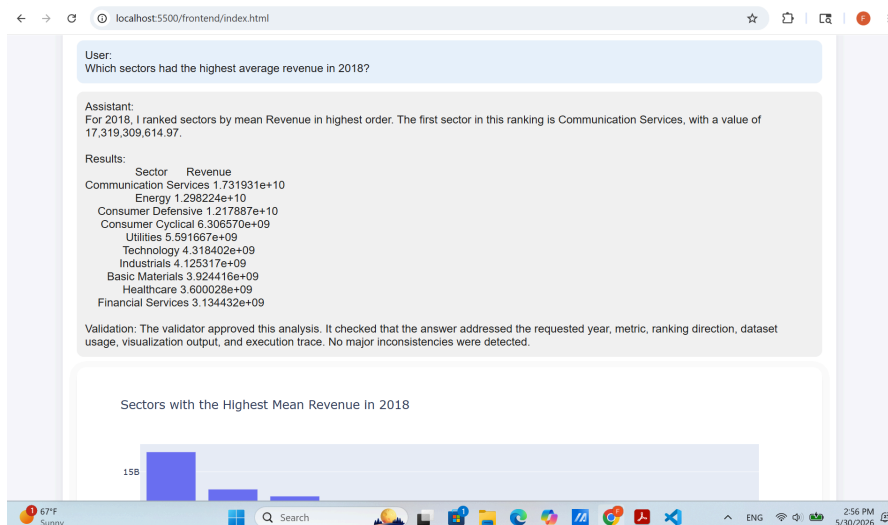
```
[{"step": "query_rewrite", "original_question": "What is retrieval-augmented generation?", "query_variants_used": [{"query": "What is retrieval-augmented generation?"}], "step": "duckduckgo_search", "num_results": 0, "first_result": null}, {"step": "answer_synthesis", "observation": "Generated a direct answer using LLM synthesis when available."}]
```

Screenshot 3: Query 3
User: What is NVIDIA's primary business segment?
Assistant: NVIDIA's primary business segment is Graphics Processing Units (GPUs) for gaming and professional markets, along with a significant focus on artificial intelligence (AI) computing.
Sources: [NVIDIA Official Site](http://www.nvidia.com/page/home.html), [NVIDIA Wikipedia](https://en.wikipedia.org/wiki/Nvidia)
Execution / Validation Trace:

```
[{"step": "query_rewrite", "original_question": "What is NVIDIA's primary business segment?", "query_variants_used": [{"query": "What is NVIDIA's primary business segment?"}], "step": "duckduckgo_search", "num_results": 5, "first_result": "Nvidia - Official Site"}, {"step": "answer_synthesis", "observation": "Generated a direct answer using LLM synthesis when available."}]
```

Next 2 examples: validator approved analytic questions:

1. Which sectors had the highest average revenue in 2018?
2. Which sector had the lowest net income in 2017?



```

{
  "iteration": 2,
  "action": "sector_ranking",
  "observation": {
    "year": "2018",
    "group_by": "Sector",
    "metric": "Revenue",
    "aggregation": "mean",
    "direction": "highest"
  },
  },
  {
    "iteration": 3,
    "action": "generate_plot",
    "observation": "Generated sector ranking bar chart."
  },
  {
    "iteration": 4,
    "action": "finalize_answer",
    "observation": "Prepared sector ranking answer."
  },
  {
    "iteration": "validation",
    "action": "validator_check",
    "observation": {
      "status": "passed",
      "passed": true,
      "summary": "Validation passed. The answer appears consistent with the question, uses appropriate data, and includes sufficient explanation and visualization.",
      "issues": [],
      "suggestions": [],
      "checks_passed": [
        "The analytics agent completed execution successfully.",
        "The answer includes a result summary or table.",
        "Detected requested year(s): ['2018'].",
        "The answer appears to address the requested metric: revenue.",
        "The answer direction appears consistent with a highest/strongest request.",
        "At least one Plotly visualization was generated.",
        "The execution trace contains multiple reasoning/execution steps."
      ],
      "validation_paragraph": "Validation: The validator approved this analysis. It checked that the answer addressed the requested year, metric, ranking direction, dataset usage, visualization output, and execution trace. No major inconsistencies were detected."
    }
  }
}

```

User:
Which sectors had the lowest average net income in 2017?

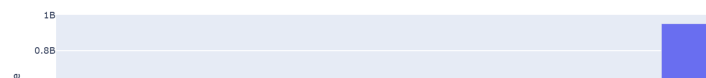
Assistant:
For 2017, I ranked sectors by mean Net Income in lowest order. The first sector in this ranking is Real Estate, with a value of 165,944,352.14.

Results:

Sector	Net Income
Real Estate	1.659444e+08
Healthcare	1.679147e+08
Industrials	2.732028e+08
Basic Materials	2.803441e+08
Consumer Cyclical	3.670619e+08
Technology	3.684931e+08
Energy	4.090049e+08
Utilities	4.173321e+08
Financial Services	4.528595e+08
Consumer Defensive	9.505049e+08

Validation: The validator approved this analysis. It checked that the answer addressed the requested year, metric, ranking direction, dataset usage, visualization output, and execution trace. No major inconsistencies were detected.

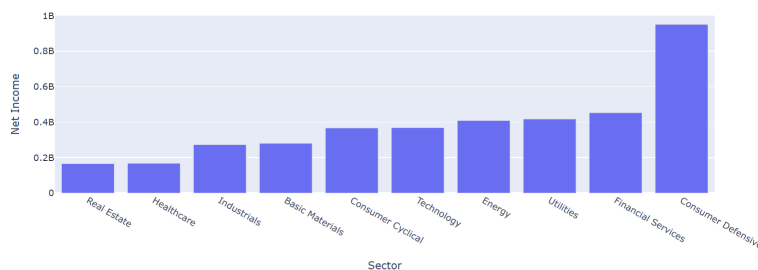
Sectors with the Lowest Mean Net Income in 2017



Utilities 4.173321e+08
Financial Services 4.528595e+08
Consumer Defensive 9.505049e+08

Validation: The validator approved this analysis. It checked that the answer addressed the requested year, metric, ranking direction, dataset usage, visualization output, and execution trace. No major inconsistencies were detected.

Sectors with the Lowest Mean Net Income in 2017



► Execution / Validation Trace

Next 2 Examples: validator did not fully approve and requested a retry

- 1. Which sectors had the lowest average revenue in 2018
- 2. Find undervalued technology companies from the dataset

User:
Which sectors appeared financially weakest from 2014 to 2018?

Assistant:
The financially weakest sectors from 2014 to 2018 were ranked using a composite score. The score rewards stronger revenue, net income, free cash flow, ROE, ROIC, and liquidity, and penalizes higher debt-to-equity.

Results:

Sector	Revenue	Net Income	Free Cash Flow	ROE	ROIC	Current ratio	Debt to Equity	Financial Strength Score
Basic Materials	3.590118e+09	1.460942e+08	1.925813e+08	0.007149	-0.363965	5.858320	0.589187	0.350649
Healthcare	3.269323e+09	2.026022e+08	2.942114e+08	-10.357858	-2.528024	6.248392	0.079423	0.376623
Real Estate	8.853335e+08	1.522529e+08	5.427497e+07	0.083213	0.075544	4.792590	1.340585	0.376623
Utilities	5.327503e+09	4.380489e+08	1.942583e+07	0.055615	0.060771	1.286011	1.018072	0.415584
Industrials	3.857562e+09	2.583931e+08	2.364818e+08	0.232790	-0.040282	2.172311	0.797641	0.441558
Consumer Cyclical	5.827106e+09	3.828283e+08	3.903640e+08	0.134663	0.047297	1.923717	1.640578	0.506494
Energy	1.202987e+10	2.799709e+08	3.042205e+08	-0.945839	1.720770	2.211717	0.756852	0.558442
Technology	3.618623e+09	4.100700e+08	5.245162e+08	0.419496	-0.135252	3.076660	0.059237	0.610390
Consumer Defensive	1.202750e+10	7.606962e+08	7.662137e+08	-0.070186	-0.038801	2.056705	0.016057	0.636364
Communication Services	1.841961e+10	1.374047e+09	1.406867e+09	0.104610	-0.012889	1.505702	-0.909850	0.753247

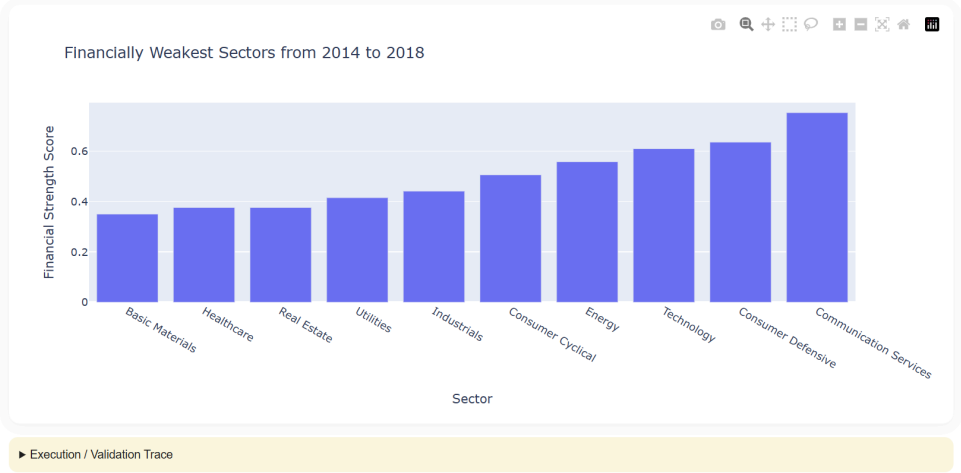
Validation: The validator did not fully approve this analysis. Main concern(s): The answer does not clearly describe change or growth between 2014 and 2018. The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is not well demonstrated. Recommended correction(s): Compute and report absolute change and percentage growth. Include inspect, execute, observe, plot, and finalize steps in the trace.

Retry Note: The validator requested a retry because the first analysis was incomplete or suspicious. The system reran the analytics agent using the same structured plan.

Financially Weakest Sectors from 2014 to 2018

2014 and 2018. The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is not well demonstrated. Recommended correction(s): Compute and report absolute change and percentage growth. Include inspect, execute, observe, plot, and finalize steps in the trace.

Retry Note: The validator requested a retry because the first analysis was incomplete or suspicious. The system reran the analytics agent using the same structured plan.



Execution/validation Trace:

```
[
  {
    "iteration": "planner",
    "action": "llm_plan",
    "observation": {
      "route": "analytics",
      "analysis_type": "financial_strength",
      "year": null,
      "start_year": "2014",
      "end_year": "2018",

```



```

    "group_by": "Sector",
    "metric": null,
    "aggregation": null,
    "sort_order": "ascending",
    "top_n": 10,
    "filters": {},
    "plot_type": null,
    "needs_web": false,
    "reason": "User is asking for a financial analysis of sectors over a range of years."
  }
},
{
  "iteration": 1,
  "action": "inspect_datasets",
  "observation": {
    "files": [
      "2014_Financial_Data.csv",
      "2015_Financial_Data.csv",
      "2016_Financial_Data.csv",
      "2017_Financial_Data.csv",
      "2018_Financial_Data.csv"
    ],
    "tables": {
      "2014": {
        "rows": 3808,
        "cols": 225
      },
      "2015": {
        "rows": 4120,
        "cols": 225
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      "2016": {
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      "2017": {
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        "cols": 225
      },
      "2018": {
        "rows": 4392,
        "cols": 225
      }
    }
  }
},
{
  "iteration": 2,
  "action": "financial_strength_score",

```

```

    "observation": "Computed weakest sector ranking using multi-metric financial strength score."
  },
  {
    "iteration": "validation",
    "action": "validator_check",
    "observation": {
      "status": "retry",
      "passed": false,
      "summary": "Validation failed. The validator detected incomplete or suspicious analysis. A
retry is recommended before accepting this answer.",
      "issues": [
        "The answer does not clearly describe change or growth between 2014 and 2018.",
        "The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is
not well demonstrated."
      ],
      "suggestions": [
        "Compute and report absolute change and percentage growth.",
        "Include inspect, execute, observe, plot, and finalize steps in the trace."
      ],
      "checks_passed": [
        "The analytics agent completed execution successfully.",
        "The answer includes a result summary or table.",
        "Detected requested year(s): ['2014', '2018'].",
        "Cross-year analysis used multiple yearly files.",
        "The answer direction appears consistent with a lowest/weakest request.",
        "Financial strength answer includes multi-metric reasoning.",
        "At least one Plotly visualization was generated."
      ],
      "validation_paragraph": "Validation: The validator did not fully approve this analysis. Main
concern(s): The answer does not clearly describe change or growth between 2014 and 2018. The
execution trace has fewer than 3 steps, so the iterative reasoning/execution process is not well
demonstrated. Recommended correction(s): Compute and report absolute change and
percentage growth. Include inspect, execute, observe, plot, and finalize steps in the trace."
    }
  },
  {
    "iteration": "planner_retry",
    "action": "reuse_llm_plan",
    "observation": {
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      "analysis_type": "financial_strength",
      "year": null,
      "start_year": "2014",
      "end_year": "2018",
      "group_by": "Sector",
      "metric": null,
      "aggregation": null,
      "sort_order": "ascending",
      "top_n": 10,

```

```

    "filters": {},
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    "needs_web": false,
    "reason": "User is asking for a financial analysis of sectors over a range of years."
  }
},
{
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  "action": "inspect_datasets",
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    "files": [
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      "2015_Financial_Data.csv",
      "2016_Financial_Data.csv",
      "2017_Financial_Data.csv",
      "2018_Financial_Data.csv"
    ],
    "tables": {
      "2014": {
        "rows": 3808,
        "cols": 225
      },
      "2015": {
        "rows": 4120,
        "cols": 225
      },
      "2016": {
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        "cols": 225
      },
      "2017": {
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      "2018": {
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        "cols": 225
      }
    }
  }
},
{
  "iteration": 2,
  "action": "financial_strength_score",
  "observation": "Computed weakest sector ranking using multi-metric financial strength score."
},
{
  "iteration": "validation_retry",
  "action": "validator_check_after_retry",

```

```

"observation": {
  "status": "retry",
  "passed": false,
  "summary": "Validation failed. The validator detected incomplete or suspicious analysis. A
retry is recommended before accepting this answer.",
  "issues": [
    "The answer does not clearly describe change or growth between 2014 and 2018.",
    "The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is
not well demonstrated."
  ],
  "suggestions": [
    "Compute and report absolute change and percentage growth.",
    "Include inspect, execute, observe, plot, and finalize steps in the trace."
  ],
  "checks_passed": [
    "The analytics agent completed execution successfully.",
    "The answer includes a result summary or table.",
    "Detected requested year(s): ['2014', '2018'].",
    "Cross-year analysis used multiple yearly files.",
    "The answer direction appears consistent with a lowest/weakest request.",
    "Financial strength answer includes multi-metric reasoning.",
    "At least one Plotly visualization was generated."
  ],
  "validation_paragraph": "Validation: The validator did not fully approve this analysis. Main
concern(s): The answer does not clearly describe change or growth between 2014 and 2018. The
execution trace has fewer than 3 steps, so the iterative reasoning/execution process is not well
demonstrated. Recommended correction(s): Compute and report absolute change and
percentage growth. Include inspect, execute, observe, plot, and finalize steps in the trace."
}
}
]

```

User:
Find undervalued technology companies from the dataset.

Assistant:
In 2018, I screened for undervalued companies using low valuation ratios, positive free cash flow, manageable debt, positive net income, and stronger ROE.

Results:
No candidates matched all filters.

Validation: The validator did not fully approve this analysis. Main concern(s): The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is not well demonstrated. Recommended correction(s): Include inspect, execute, observe, plot, and finalize steps in the trace.

Retry Note: The validator requested a retry because the first analysis was incomplete or suspicious. The system reran the analytics agent using the same structured plan.

► Execution / Validation Trace

Execution/Validation Trace:

```

[
  {
    "iteration": "planner",
    "action": "llm_plan",
    "observation": {

```

```

    "route": "analytics",
    "analysis_type": "undervalued_screen",
    "year": null,
    "start_year": null,
    "end_year": null,
    "group_by": "Sector",
    "metric": null,
    "aggregation": null,
    "sort_order": null,
    "top_n": 10,
    "filters": {
      "Sector": "Technology"
    },
    "plot_type": null,
    "needs_web": false,
    "reason": "User is looking for undervalued companies specifically in the technology sector
from the dataset."
  }
},
{
  "iteration": 1,
  "action": "inspect_datasets",
  "observation": {
    "files": [
      "2014_Financial_Data.csv",
      "2015_Financial_Data.csv",
      "2016_Financial_Data.csv",
      "2017_Financial_Data.csv",
      "2018_Financial_Data.csv"
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    "tables": {
      "2014": {
        "rows": 3808,
        "cols": 225
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      "2015": {
        "rows": 4120,
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      },
      "2017": {
        "rows": 4960,
        "cols": 225
      },
      "2018": {
        "rows": 4392,

```

```

        "cols": 225
      }
    }
  },
  {
    "iteration": 2,
    "action": "undervalued_screen",
    "observation": "Found 0 candidates in 2018."
  },
  {
    "iteration": "validation",
    "action": "validator_check",
    "observation": {
      "status": "retry",
      "passed": false,
      "summary": "Validation failed. The validator detected incomplete or suspicious analysis. A
retry is recommended before accepting this answer.",
      "issues": [
        "The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is
not well demonstrated."
      ],
      "suggestions": [
        "Include inspect, execute, observe, plot, and finalize steps in the trace."
      ],
      "checks_passed": [
        "The analytics agent completed execution successfully.",
        "The answer includes a result summary or table."
      ],
      "validation_paragraph": "Validation: The validator did not fully approve this analysis. Main
concern(s): The execution trace has fewer than 3 steps, so the iterative reasoning/execution
process is not well demonstrated. Recommended correction(s): Include inspect, execute,
observe, plot, and finalize steps in the trace."
    }
  },
  {
    "iteration": "planner_retry",
    "action": "reuse_llm_plan",
    "observation": {
      "route": "analytics",
      "analysis_type": "undervalued_screen",
      "year": null,
      "start_year": null,
      "end_year": null,
      "group_by": "Sector",
      "metric": null,
      "aggregation": null,
      "sort_order": null,
      "top_n": 10,

```

```

    "filters": {
      "Sector": "Technology"
    },
    "plot_type": null,
    "needs_web": false,
    "reason": "User is looking for undervalued companies specifically in the technology sector
from the dataset."
  }
},
{
  "iteration": 1,
  "action": "inspect_datasets",
  "observation": {
    "files": [
      "2014_Financial_Data.csv",
      "2015_Financial_Data.csv",
      "2016_Financial_Data.csv",
      "2017_Financial_Data.csv",
      "2018_Financial_Data.csv"
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    "tables": {
      "2014": {
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      "2015": {
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      "2016": {
        "rows": 4797,
        "cols": 225
      },
      "2017": {
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      "2018": {
        "rows": 4392,
        "cols": 225
      }
    }
  }
},
{
  "iteration": 2,
  "action": "undervalued_screen",
  "observation": "Found 0 candidates in 2018."
},

```

```

{
  "iteration": "validation_retry",
  "action": "validator_check_after_retry",
  "observation": {
    "status": "retry",
    "passed": false,
    "summary": "Validation failed. The validator detected incomplete or suspicious analysis. A
retry is recommended before accepting this answer.",
    "issues": [
      "The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is
not well demonstrated."
    ],
    "suggestions": [
      "Include inspect, execute, observe, plot, and finalize steps in the trace."
    ],
    "checks_passed": [
      "The analytics agent completed execution successfully.",
      "The answer includes a result summary or table."
    ],
    "validation_paragraph": "Validation: The validator did not fully approve this analysis. Main
concern(s): The execution trace has fewer than 3 steps, so the iterative reasoning/execution
process is not well demonstrated. Recommended correction(s): Include inspect, execute,
observe, plot, and finalize steps in the trace."
  }
}
]

```

8. Main Scenario: Interactions

FinSight Orchestrator
Multi-agent AI analytics system for financial stock indicators

User:
Which sectors experienced the largest improvement in net income between 2014 and 2018?

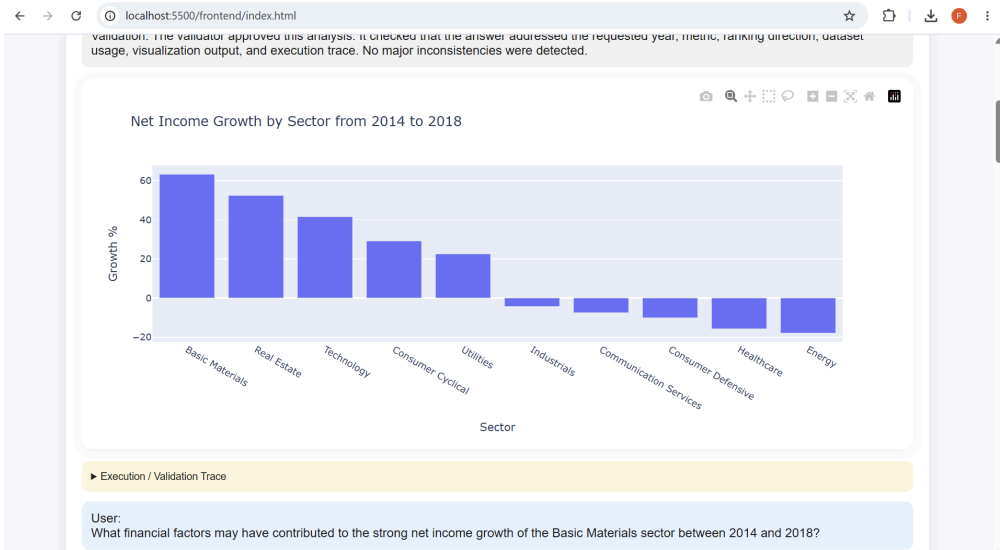
Assistant:
I compared average sector Net Income from 2014 to 2018. The sector with the highest growth in this ranking is Basic Materials, with growth of 63.24%.

Results:

Sector	Net Income_2014	Net Income_2018	Change	Growth %
Basic Materials	1.904975e+08	3.109650e+08	1.204675e+08	63.238382
Real Estate	1.169704e+08	1.782985e+08	6.132808e+07	52.430416
Technology	3.806563e+08	5.387657e+08	1.581094e+08	41.535999
Consumer Cyclical	3.383290e+08	4.369290e+08	9.860001e+07	29.143235
Utilities	4.419788e+08	5.414921e+08	9.951327e+07	22.515394
Industrials	2.773852e+08	2.653626e+08	-1.202260e+07	-4.334263
Communication Services	1.530762e+09	1.416082e+09	-1.146800e+08	-7.491699
Consumer Defensive	7.786109e+08	6.998263e+08	-7.878465e+07	-10.118616
Healthcare	2.247882e+08	1.893828e+08	-3.540534e+07	-15.750537
Energy	7.945380e+08	6.519805e+08	-1.425576e+08	-17.942199

Validation: The validator approved this analysis. It checked that the answer addressed the requested year, metric, ranking direction, dataset usage, visualization output, and execution trace. No major inconsistencies were detected.

Net Income Growth by Sector from 2014 to 2018



localhost:5500/frontend/index.html

User:
What financial factors may have contributed to the strong net income growth of the Basic Materials sector between 2014 and 2018?

Assistant:
The financially strongest sectors from 2014 to 2018 were ranked using a composite score. The score rewards stronger revenue, net income, free cash flow, ROE, ROIC, and liquidity, and penalizes higher debt-to-equity.

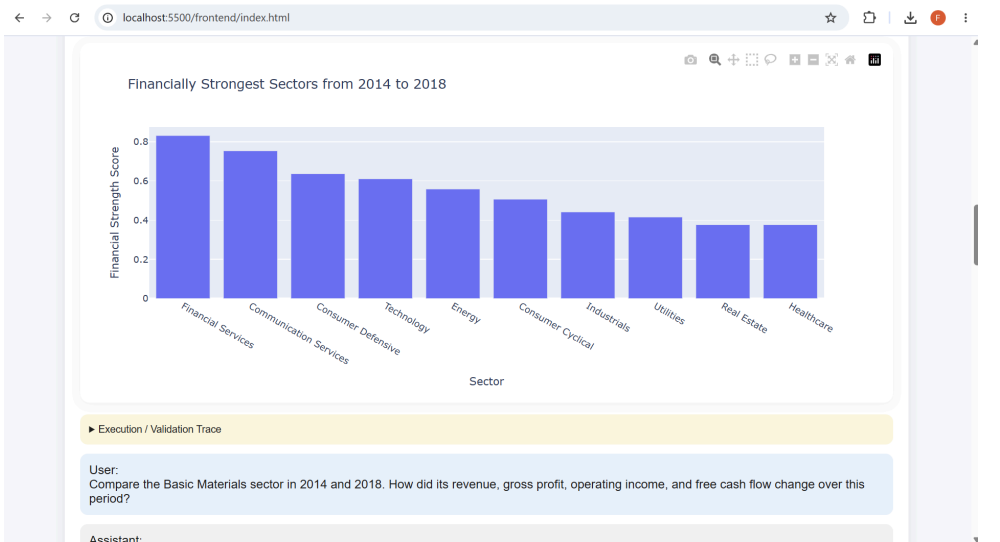
Results:

Sector	Revenue	Net Income	Free Cash Flow	ROE	ROIC	Current ratio	Debt to Equity	Financial Strength Score
Financial Services	5.214934e+09	6.414431e+08	9.190649e+08	10073.251616	15.828258	162.921566	0.448854	0.831169
Communication Services	1.841961e+10	1.374047e+09	1.406867e+09	0.104610	-0.012889	1.505702	-0.909850	0.753247
Consumer Defensive	1.202750e+10	7.606962e+08	7.662137e+08	-0.070186	-0.038801	2.056705	0.016057	0.636364
Technology	3.618623e+09	4.100700e+08	5.245162e+08	0.419496	-0.135252	3.076660	0.059237	0.610390
Energy	1.202987e+10	2.799709e+08	3.042205e+08	-0.945839	1.720770	2.211717	0.756852	0.558442
Consumer Cyclical	5.827106e+09	3.828283e+08	3.903640e+08	0.134663	0.047297	1.923717	1.640578	0.506494
Industrials	3.857562e+09	2.583931e+08	2.364819e+08	0.232790	-0.040282	2.172311	0.797641	0.441558
Utilities	5.327503e+09	4.380489e+08	1.942583e+07	0.055615	0.060771	1.286011	1.018072	0.415584
Real Estate	8.853335e+08	1.522529e+08	5.427497e+07	0.083213	0.075544	4.792590	1.340585	0.376623
Healthcare	3.269323e+09	2.026022e+08	2.942114e+08	-10.357858	-2.528024	6.248392	0.079423	0.376623

Validation: The validator did not fully approve this analysis. Main concern(s): The answer does not clearly describe change or growth between 2014 and 2018. The execution trace has fewer than 3 steps, so the iterative reasoning/execution process is not well demonstrated. Recommended correction(s): Compute and report absolute change and percentage growth. Include inspect, execute, observe, plot, and finalize steps in the trace.

Retry Note: The validator requested a retry because the first analysis was incomplete or suspicious. The system reran the analytics agent using the same structured plan.

Financially Strongest Sectors from 2014 to 2018



← → ↺ 📄 localhost:5500/frontend/index.html ⭐ 📄 📄 📄 📄

▶ Execution / Validation Trace

User:

Compare the Basic Materials sector in 2014 and 2018. How did its revenue, gross profit, operating income, and free cash flow change over this period?

Assistant:

I compared average sector Revenue from 2014 to 2018. The sector with the lowest growth in this ranking is Financial Services, with growth of -52.98%.

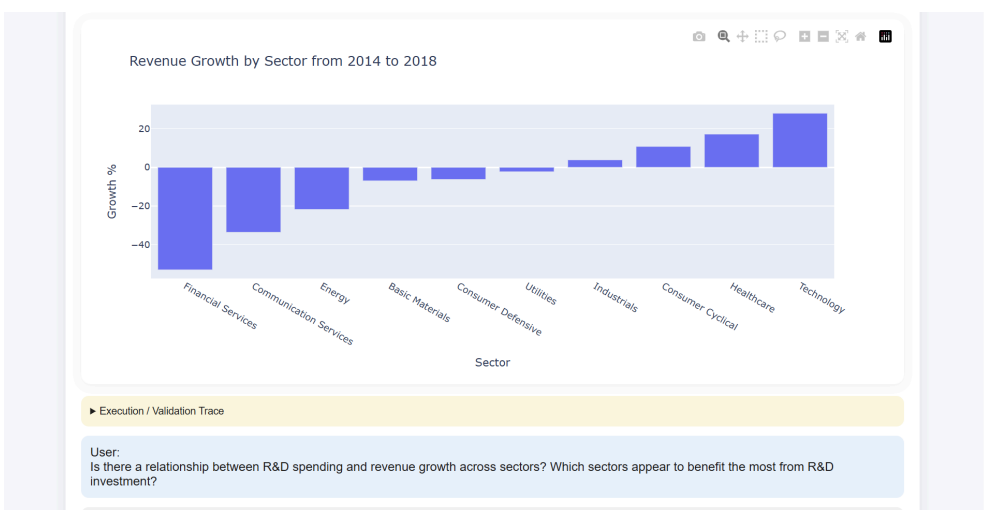
Results:

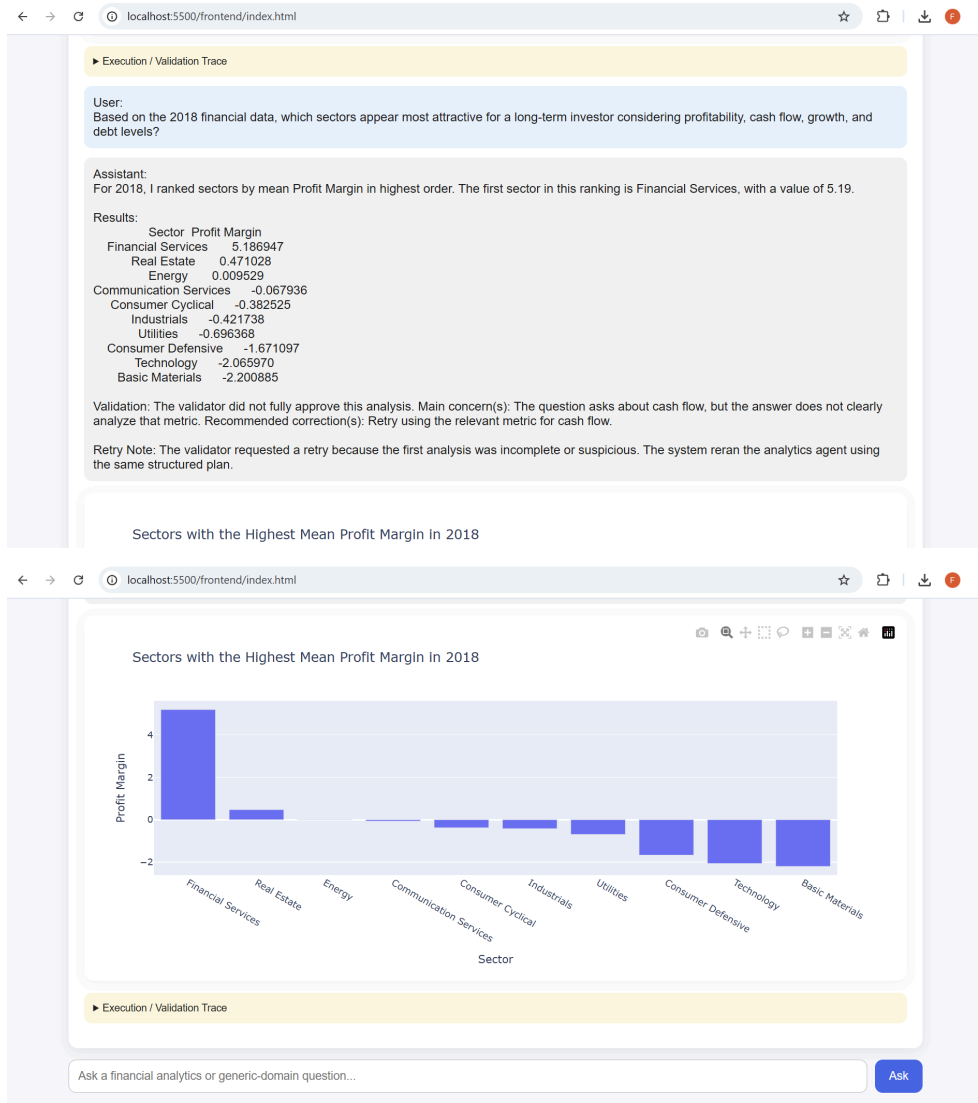
Sector	Revenue_2014	Revenue_2018	Change	Growth %
Financial Services	6.665841e+09	3.134432e+09	-3.531409e+09	-52.977697
Communication Services	2.608160e+10	1.731931e+10	-8.762292e+09	-33.595682
Energy	1.659526e+10	1.298224e+10	-3.613020e+09	-21.771391
Basic Materials	4.217512e+09	3.924416e+09	-2.930963e+08	-6.949507
Consumer Defensive	1.299110e+10	1.217887e+10	-8.122289e+08	-6.252193
Utilities	5.721887e+09	5.591667e+09	-1.302196e+08	-2.275616
Industrials	3.973373e+09	4.125317e+09	1.519441e+08	3.824059
Consumer Cyclical	5.692990e+09	6.306570e+09	6.135805e+08	10.777826
Healthcare	3.072740e+09	3.60028e+09	5.27288e+08	17.160216
Technology	3.375641e+09	4.318402e+09	9.427616e+08	27.928375

Validation:

The validator approved this analysis. It checked that the answer addressed the requested year, metric, ranking direction, dataset usage, visualization output, and execution trace. No major inconsistencies were detected.

Revenue Growth by Sector from 2014 to 2018





Findings:

- **Basic Materials** showed the largest net income growth from 2014 to 2018, increasing by approximately **63%**.
- Several follow-up analyses revealed that the analytics agent sometimes focused on the wrong metric or answered a different question than the one asked.
- The **Validator Agent successfully identified incomplete analyses**, including missing metrics, incorrect reasoning, and insufficient support for conclusions.
- Correlation analysis found **little to no relationship between R&D spending and revenue growth** at the sector level in 2018.
- Investment attractiveness assessments require multiple factors (profitability, cash flow, growth, and debt), and the validator correctly rejected analyses that relied on only a single metric.
- Overall, the multi-agent architecture demonstrated the value of **independent validation**, improving reliability by detecting analytical errors that would otherwise appear as valid answers.

9.Reasoning/Execution Iteration Explained

a. Example 1

Outputs:

User:

Which sectors experienced the largest improvement in free cash flow between 2014 and 2018?

Assistant:

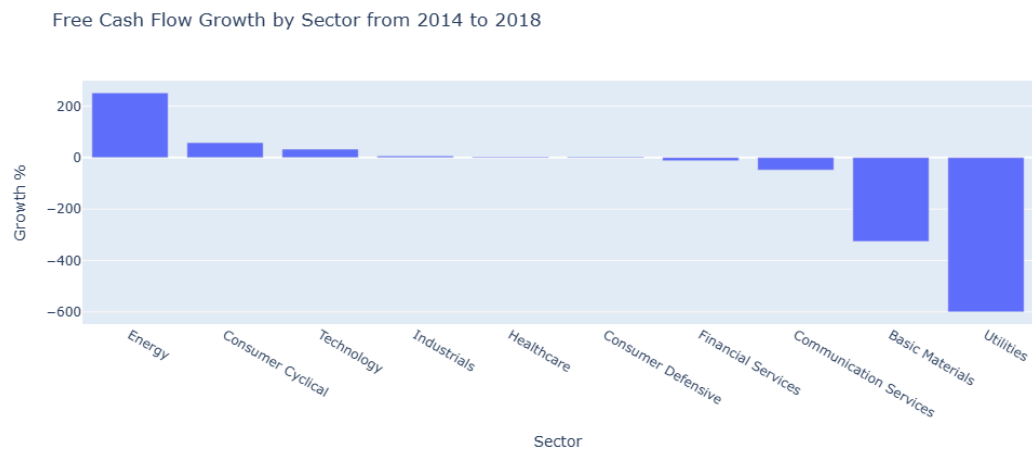
I compared average sector Free Cash Flow from 2014 to 2018. The sector with the highest growth in this ranking is Energy, with growth of 251.78%.

Results:

Change	Sector	Free Cash Flow_2014	Free Cash Flow_2018	Growth %
251.775168	Energy	1.861469e+08	6.548185e+08	4.686716e+08
1.760863e+08	Consumer Cyclical	3.040035e+08	4.800898e+08	57.922466
1.531878e+08	Technology	4.688831e+08	6.220708e+08	32.670778
6.937280	Industrials	2.263370e+08	2.420386e+08	1.570163e+07
3.693752	Healthcare	2.998040e+08	3.108780e+08	1.107402e+07
2.869044e+07	Consumer Defensive	7.945586e+08	8.232491e+08	3.610865
-1.210547e+08	Financial Services	9.906988e+08	8.696441e+08	-12.219120
-1.311240e+09	Communication Services	2.714802e+09	1.403561e+09	-48.299673
5.040389e+08	Basic Materials	-1.547119e+08	3.493270e+08	-325.792010
-600.003460	Utilities	1.665699e+07	-8.328551e+07	-9.994249e+07

Validation: The validator approved this analysis. It checked that the answer addressed the requested year, metric, ranking direction, dataset

usage, visualization output, and execution trace. No major inconsistencies were detected.



Analysis:

Question:

Which sectors experienced the largest improvement in free cash flow between 2014 and 2018?

Iteration 1: Inspect Datasets

The Analytics Agent first inspected the available financial datasets and identified the yearly CSV files from 2014 to 2018. It confirmed that both the 2014 and 2018 datasets were available and contained the required **Sector** and **Free Cash Flow** columns.

Observation:

The necessary data existed for a cross-year comparison of sector-level free cash flow.

Next Step:

Proceed with aggregating and comparing free cash flow across sectors.

Iteration 2: Perform Cross-Year Comparison

The agent grouped companies by **Sector** and computed the average **Free Cash Flow** for both 2014 and 2018. It then calculated the absolute change and percentage growth for each sector.

Observation:

Several sectors showed significant changes over time. The **Energy** sector exhibited the largest positive growth, increasing from approximately \$186 million to \$655 million in average free cash flow.

Next Step:

Rank sectors by growth percentage and prepare a visualization.

Iteration 3: Generate Results and Visualization

The agent ranked sectors according to free cash flow growth and created a Plotly bar chart showing the percentage change from 2014 to 2018.

Observation:

The visualization clearly showed that:

- **Energy** experienced the strongest improvement (+251.8%).
- **Consumer Cyclical** and **Technology** also showed substantial growth.
- **Communication Services**, **Basic Materials**, and **Utilities** experienced declines.

Next Step:

Generate the final explanation and present the results to the user.

Final Findings

The analysis found that the **Energy sector** experienced the largest improvement in average free cash flow between 2014 and 2018, with growth of approximately **251.8%**. Consumer Cyclical and Technology also demonstrated strong improvements, while Utilities and Communication Services showed substantial declines.

b. Example 2

Outputs:

User:

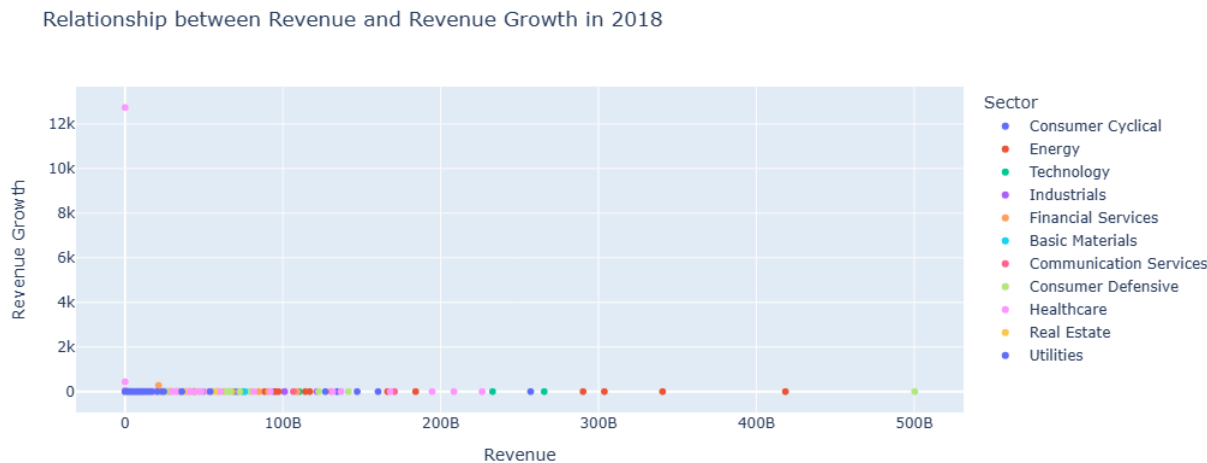
Do companies with higher debt-to-equity ratios tend to have lower ROE in 2018?

Assistant:

In 2018, the correlation between Revenue and Revenue Growth is -0.004. This suggests a weak negative relationship.

Validation: The validator did not fully approve this analysis. Main concern(s): The final answer is too short and may not sufficiently explain the result. The answer does not appear to include a result table or ranked summary. The question asks about roe, but the answer does not clearly analyze that metric. Recommended correction(s): Add a clear interpretation, metric name, year, and top results. Include a table or concise ranked summary in the final answer.

Retry Note: The validator requested a retry because the first analysis was incomplete or suspicious. The system reran the analytics agent using the same structured plan.



Analysis:

Question:

Do companies with higher debt-to-equity ratios tend to have lower ROE in 2018?

Iteration 1: Inspect Dataset

The Analytics Agent inspected the available financial datasets and identified the 2018 dataset as the correct source because the question explicitly referenced 2018. The planner determined that the task required a **correlation analysis** between **Debt-to-Equity Ratio** and **ROE**.

Observation:

The required metrics were available in the dataset, and a scatter plot was selected as the appropriate visualization method.

Next Step:

Perform correlation analysis between the selected financial metrics.

Iteration 2: Execute Correlation Analysis

The agent attempted to compute a correlation; however, it mistakenly analyzed **Revenue** and **Revenue Growth** instead of **Debt-to-Equity Ratio** and **ROE**. The resulting correlation coefficient was **-0.004**, indicating essentially no relationship.

Observation:

The computed metrics did not match the original user request. Although a correlation was calculated successfully, the wrong variables were analyzed.

Next Step:

Generate a visualization and prepare an answer based on the computed results.

Iteration 3: Generate Visualization and Final Answer

A scatter plot was generated showing the relationship between **Revenue** and **Revenue Growth**, and the system concluded that the relationship was weakly negative. However, the answer did not discuss **ROE** or **Debt-to-Equity Ratio**, which were the metrics requested by the user.

Observation:

The generated answer was incomplete and addressed the wrong financial metrics.

Next Step:

Pass the results to the Validator Agent for independent review.

Validator Findings

The Validator Agent independently examined the question, execution trace, and generated answer. It detected several issues:

- The question requested analysis of **Debt-to-Equity Ratio** and **ROE**, but the analytics agent analyzed **Revenue** and **Revenue Growth** instead.

- The final answer was too short and lacked detailed interpretation.
- No result table or ranked summary was provided.
- The reasoning process was insufficiently documented.

Validation Result

Not Approved – Retry Recommended

The validator correctly identified that the analytics agent had answered a different question than the one asked. Even though the computation executed successfully and a visualization was generated, the analysis was not relevant to the user's request.

Final Finding

This example demonstrates the importance of an independent Validator Agent. The analytics agent successfully completed an analysis, but it analyzed the wrong variables. The validator detected this semantic mismatch and prevented an incorrect result from being accepted, illustrating how validation improves the reliability and trustworthiness of the overall multi-agent system.

10. Summary:

The orchestrated AI analytics system successfully demonstrated how multiple specialized agents can collaborate to answer financial analysis questions using natural language. The system combined planning, execution, validation, memory, and web search capabilities to provide structured and explainable analytical results.

The system terminates when either a valid answer is produced and approved by the Validator Agent, or when the maximum retry limit is reached. If the validator detects incomplete reasoning, incorrect dataset usage, missing metrics, or inconsistencies between the question and the generated answer, it can request a retry. In the current implementation, a single retry attempt is performed before returning the final result.

Several validation strategies were implemented. The Validator Agent checks answer completeness, numerical consistency, correct dataset and year selection, ranking direction (e.g., highest vs. lowest), cross-year dataset usage, visualization generation, and execution trace quality. It independently compares the user's question with the generated analysis and identifies potential mismatches or missing analytical steps.

Despite these capabilities, several limitations were observed. The Planner Agent occasionally misinterpreted user intent and selected an incorrect analysis type. The Analytics Agent sometimes analyzed the wrong variables or focused on only part of a

multi-part question. In some cases, the Validator Agent successfully detected these errors, while in others it approved analyses that were technically correct but did not fully answer the user's question. These examples highlight the difficulty of validating semantic correctness in open-ended analytical tasks.

Several implementation challenges were encountered during development. These included designing a flexible planning mechanism, handling multiple yearly datasets, supporting different analysis types with a common framework, generating meaningful visualizations, and developing a validator capable of reasoning about analytical correctness rather than simply checking execution success. Integrating the planner, analytics, validator, memory, and frontend/backend components into a unified workflow was also a significant challenge.

Future improvements could include a more sophisticated LLM-based planner, enhanced semantic validation, multi-step corrective retries that modify the analysis plan, support for additional operations such as joins and advanced statistical analysis, and stronger reasoning capabilities for handling complex multi-part questions. These improvements would further increase the robustness, accuracy, and trustworthiness of the system.